

Design and function

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General structure of the machine

Base frame	The machine is mounted on a base frame made of stainless steel with vibration-damping and variable-height supports.
Panelling	The machine is equipped with protective doors with protective switches and with stainless steel doors with locking levers in the areas for drive and installation. For further information on the safety equipment, please refer to the Chapter "Safety".
Production, drive and installation areas	The production area is separated from the drive and installation areas by means of a plate. This means that, depending on the machine type, the machine's electric and pneumatic equipment is housed in the rear or lower part of the machine.

Limits of the machine



Attention!

If hazardous, flammable/inflammable materials are processed, the system owner is responsible for:

- safe processing,
- extraction device in the working area.

Fundamentals of the machine design

Principle The machine/system is constructed according to the state of technology and the recognised safety rules. Nevertheless, when used, dangers may arise for the life and limb of the user or third parties or the machine and other property may be impaired.

Using the machine The machine must only be operated in perfect technical working order and in the way intended, with attention given to safety and potential dangers whilst observing the operating instructions! In particular, immediately rectify (or have rectified) malfunctions which could affect safety!

Intended use

Application of the machine The machine is suitable for assembling tips and caps together and for sealing them along with infed catheters in film.

The machine may only be operated as intended with the products listed in the corresponding business order and/or formats. Only personnel who are appropriately qualified to perform their respective tasks may work at the machine (see operating personnel / user groups).

Any other use is considered as non-intended use. The manufacturer accepts no liability for damage resulting as a consequence of this; the system owner bears the associated risk.

The intended use includes the adherence to the operating, maintenance and repair conditions specified by the manufacturer.

Residual risk

In spite of all precautions, the following residual risk remains:



Attention!

Risk of injury.

The protective doors are bypassed in setup mode. As a result, it is possible to perform machine movements while the safety doors are open. Only authorised and trained persons who are familiar with the machine operation and are aware of the possible dangers may bridge protective doors.

- In setup mode, the machine may only be operated by one person.
- Secure the machine against accidental operation by another person, e.g. affix warning sign.

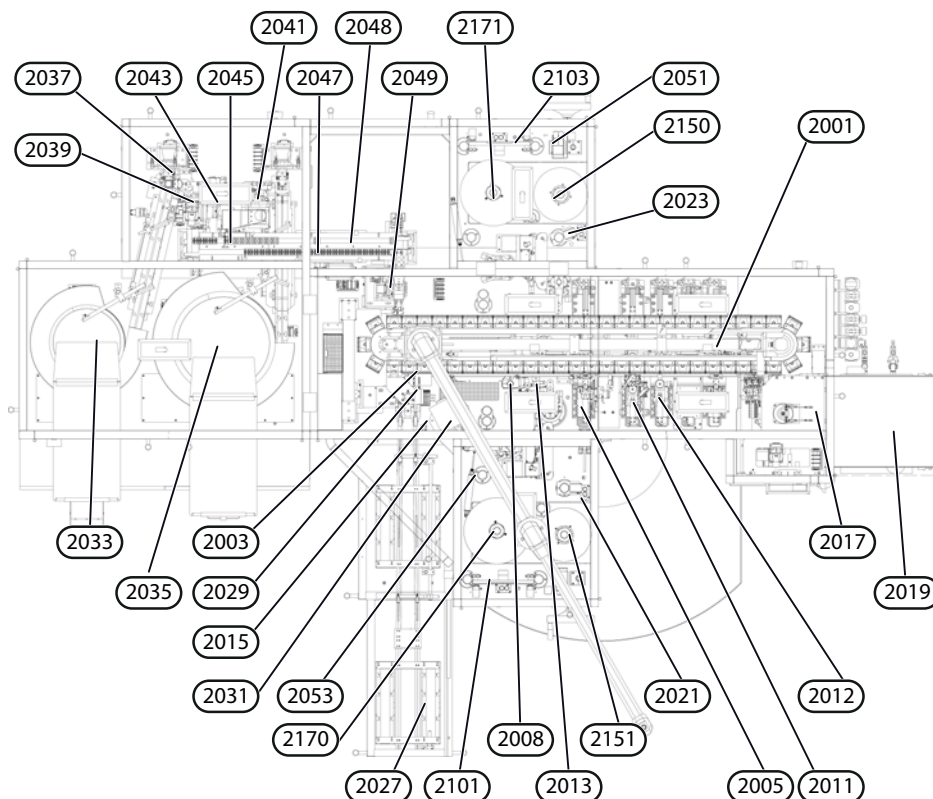


Warning!

Risk of crushing.

Tools can crush hands and fingers.
Do not reach in between the station tools.

Machine overview



Stationen.ai

The machine consists of the following stations:

Station 2001 - Oval turret

Station 2003 - Insert catheter in tips

Station 2005 - Tack sealing funnel & tip

Station 2008 - Straightness check of mandrel

Station 2011 - Long seal

Station 2012 - Foil separation

Station 2013 - Funnel & tip sealing

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Station 2015 - Pick & place catheters
Station 2017 - Reject station
Station 2019 - Discharge conveyor
Station 2021 - Splice detection front film
Station 2023 - Splice detection rear film
Station 2027 - Punched catheter feeding
Station 2029 - Punched catheter transfer
Station 2031 - CH verification
Station 2033 - Tip feeding
Station 2035 - Cap feeding
Station 2037 - Insert tip
Station 2039 - Vision system tip
Station 2041 - Insert cap
Station 2043 - Cap/tip assembly
Station 2045 - Cap/tip transfer
Station 2047 - Load pucks/buffer
Station 2048 - Unload pucks
Station 2049 - Insert cap/tip
Station 2051 - Position verification front film
Station 2053 - Position verification rear film
Station 2101 - Edge guiding front film
Station 2103 - Edge guiding rear film
Station 2150 - Unwind front film
Station 2151 - Unwind rear film
Station 2170 - Rewind front film carrier
Station 2171 - Rewind rear film carrier

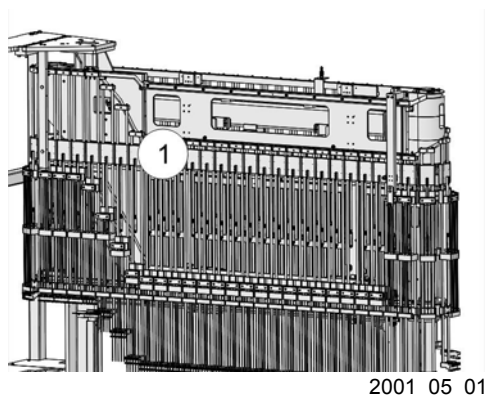
Station 2001 - Oval turret

The oval turntable is arranged vertically and mounted on the machine plate. It is driven by 3 servo-controlled toothed belts:

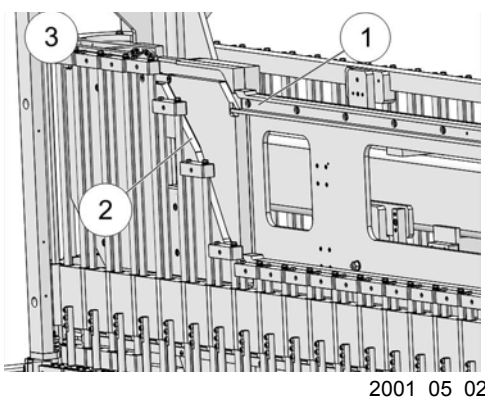
- The top belt holds the clamping plate arbor
- The middle belt holds the guide pushers for the funnel containers.
- The bottom belt holds the workpiece carriers of the caps/tip pucks.

A workpiece carrier is equipped with four receivers.

A format change between 20 cm and 40 cm is carried out by changing over pneumatically between 2 sets of cam tracks.



- Turntable (1)



- Upper cam track (1) for format 40 cm
- Lower cam track (2) for format 20 cm
- Pneumatically-switched gate lever (3)

Station 2003 - Insert catheter in tips

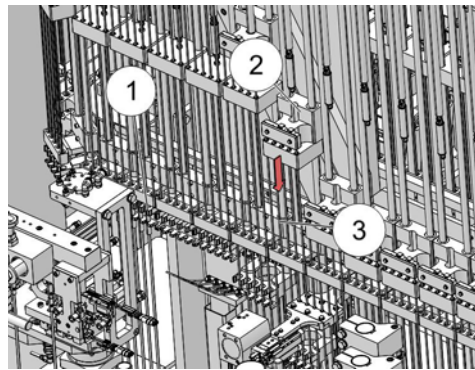
The catheters are collected from the infeed by two grippers and inserted in the receivers of the oval runner.

As the catheters are collected, they are aligned by two pins.

A special guide ensures clean infeed of the catheters into the tips.

The mandrels are lowered into the catheters and stabilise the catheter tubes as they are inserted in the cap/tip assembly group.

The guides run another two cycles whereupon they move back up again.



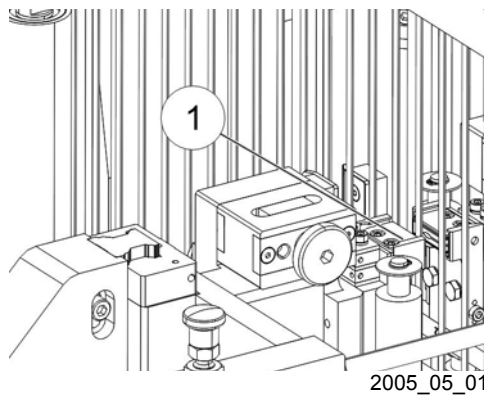
2003_05_02

- Catheter in support position (1)
- Guide tube holder (2)
- Mandrel (3)

Station 2005 - Tack sealing funnel & tip

The PU films are joined by means of hot sealing points (impulse sealing technology) to the funnel and inserting tip.

The sealing station is mounted on the base plate and is self-supporting.



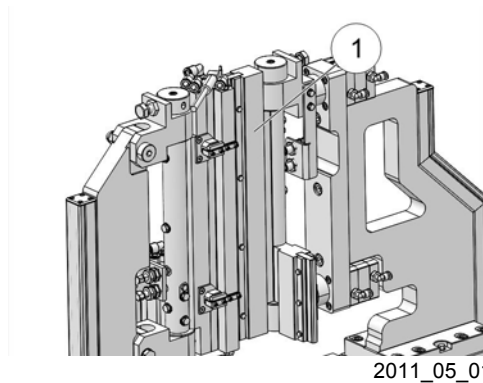
- Sealing head (1) of cycle sealing

Station 2011 - Long seal

The long sealing station seals the transverse seams across the whole width of the PU film. The station is equipped with a long (format 40 cm) and a short (format 20 cm) heated sealing tool. The station swivels 90° depending on the format.

The sealing tool features a cutting edge. The sealing tool presses the film against a rubber anvil and the cutting edge cuts through the film in the centre of the sealed surface.

The station is mounted on the base plate and is self-supporting.

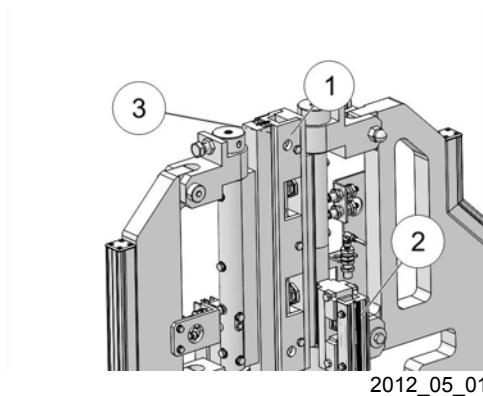


- Long sealing tool (1)

Station 2012 - Foil separation

The film is tensioned and cut through with a pneumatically-powered blade.

The station is equipped with a long (format 40 cm) and a short (format 20 cm) clamping jaw. The station swivels 90° depending on the format.

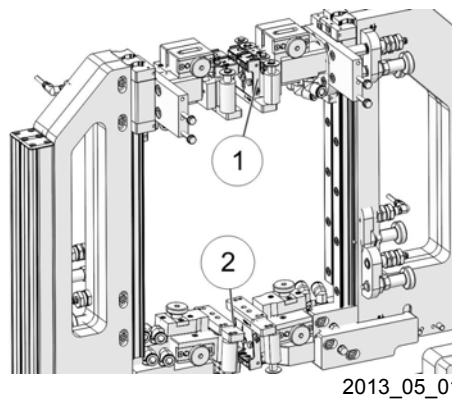


- Long clamping jaw (1) in working position
- Short clamping jaw (2) swivelled away by 90°
- Blade (3)

Station 2013 - Funnel & tip sealing

The PU films are joined to the inserting tip by means of hot sealing points (impulse sealing technology).

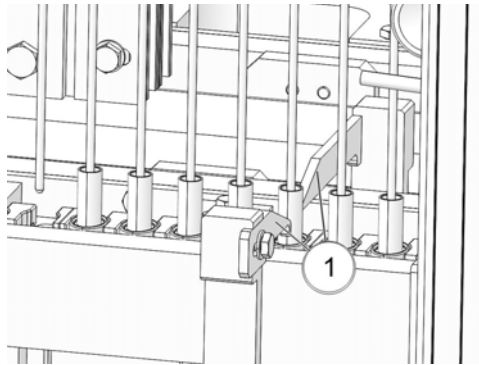
The sealing station is mounted on the base plate and is self-supporting.



- Sealing head (1) for funnel sealing
- Sealing (2) head for tip sealing

Station 2014 - Tip press in

After sealing and cutting, the station once again presses the caps against the tips with two grippers.



2014_05_01

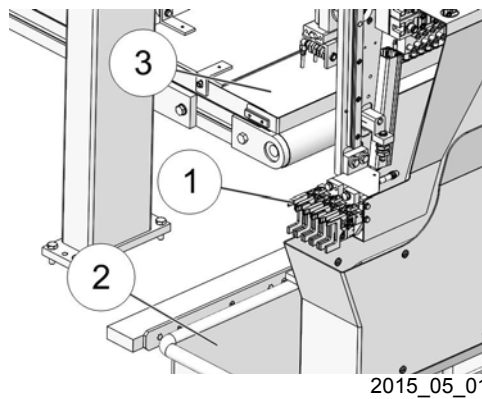
- Gripper (1)

Station 2015 - Pick & place catheters

4 catheters are picked up simultaneously with special parallel grippers. The gripper pulls the catheters out of the workpiece carrier and swivels them 90° to the horizontal position.

Bad parts are dropped here into the bad parts bin by the relevant gripper opening.

The good parts are then swivelled further with the gripper over the discharge belt and released.

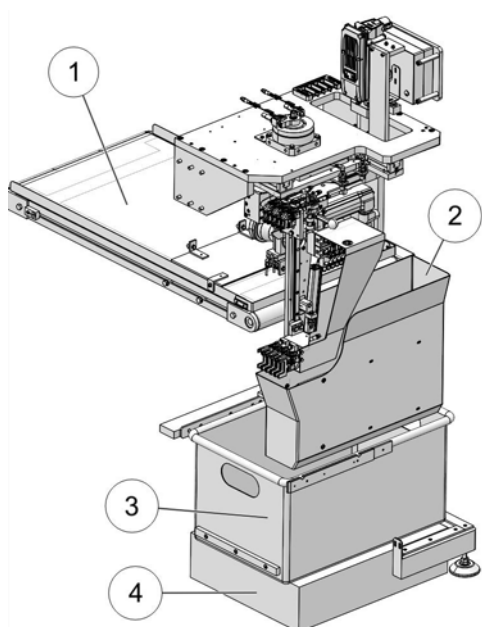


- Pick&Place parallel gripper (1)
- Bad parts bin station 2017 (2)
- Conveyor belt for good parts station 2019 (3)

Station 2017 - Reject station

Catheters that have been declared as rejects are dropped into the bad parts bin by the Pick & Place grippers of the previous station. Bad parts testing is checked with a set of scales.

The bad parts bin can be emptied during operation or while the system is stationary. The machine will continue to run with a full or missing container until the next bad part is detected.



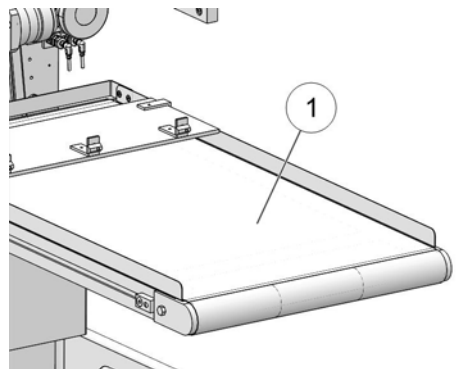
2017_05_01

- Discharge belt (1)
- Bad-part slide (2)
- Bad parts bin (3)
- Scales (4)

Station 2019 - Discharge conveyor

The good parts are placed on the conveyor belt.

The good parts are removed from the belt manually.

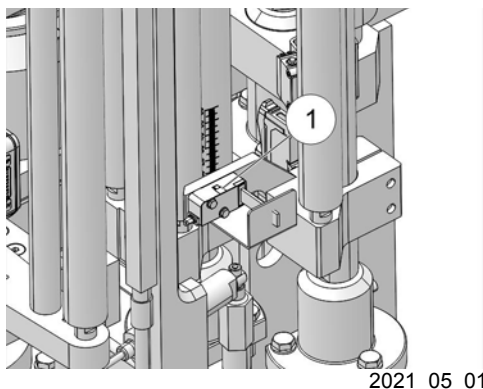


2019_05_01

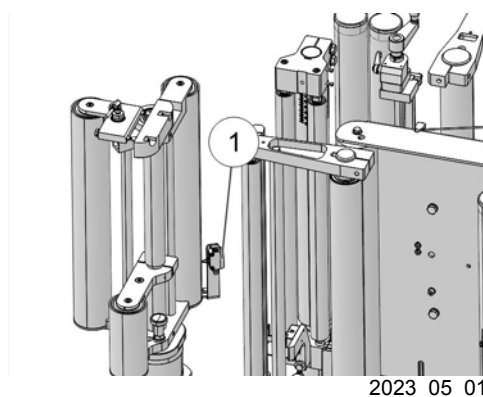
- Conveyor belt (1)

Station 2021 - Splice detection front film Station 2023 - Splice detection rear film

The splicing web detection checks the incoming film for splices on the underside of the film.



- Sensor station 2021 (1)



- Sensor station 2023 (1)

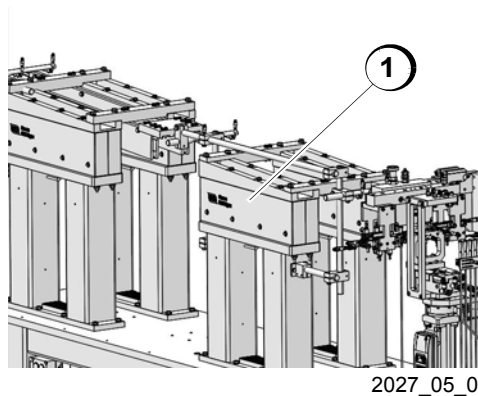
Station 2027 - Punched catheter feeding

The catheters are raised manually in two linear prefeed rails and placed on the funnel.

The prefeed rails serves as a buffer. The total (100%) buffer capacity of 240 catheters without 2 x 22 catheters between the minimum sensor monitoring and separator.

The catheters can be loaded in 3 minutes.

The infeed can be started manually by pressing a button.

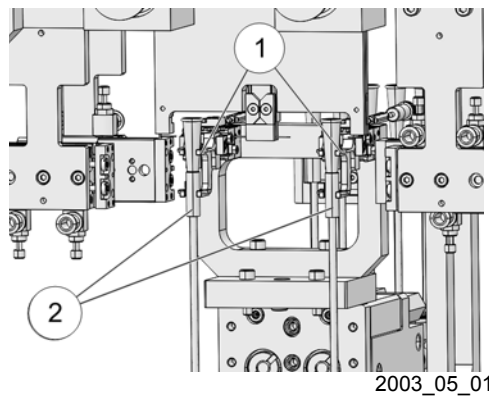


- Catheter infeed (1)

Station 2029 - Punched catheter transfer

The catheters are continually conveyed through the system in two lanes.

The opened gripper moves to the collecting position and grips two catheters. It moves back with the catheters, rotates and places the two catheters in the receivers of station 2001.

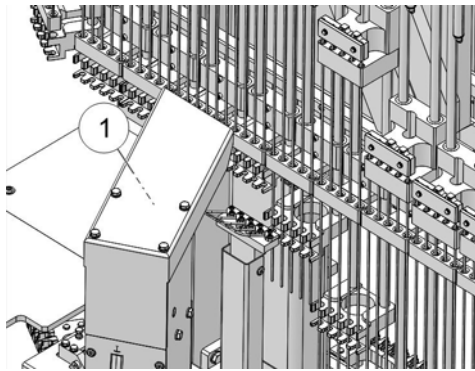


- Gripper (1)
- Catheter in gripper jaws (1)

2003_05_01

Station 2031 - CH verification

After separation in station 2029, a colour sensor (camera) checks the CH type. If the colour is incorrect, the catheter is marked as a bad part in the shift register and ejected in station 2015 into the bad parts bin of station 2017.



Colour camera (1)

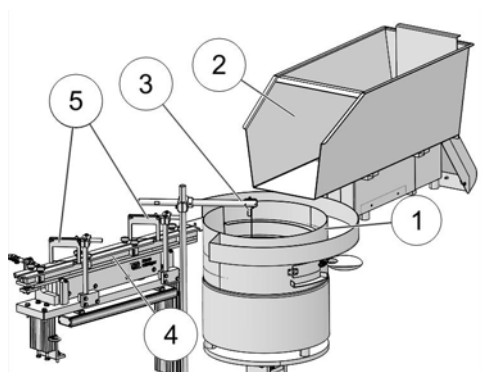
2031_05_01

Station 2033 - Tip feeding

The tip infeed consists of a container with swivel conveyor and linear conveyor as well as a discharge belt. The station transports the tips out of the spiral conveyor into the next station.

A sensor monitors the presence of the tips in the spiral conveyor. Sensors monitor the infeed of tips for maximum and minimum backlog.

Different tips can be fed in without format adjustments.



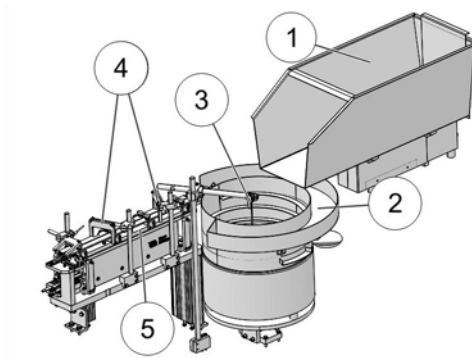
2033_05_01

- Spiral conveyor (1)
- Infeed gutter (2)
- Sensor (3)
- Discharge rails (4) for tips
- Sensors (5) for minimum and maximum backlog

Station 2035 - Cap feeding

The cap feeding unit consists of a bunker with swivel conveyor and linear conveyor as well as a discharge rail. The station transports the caps out of the spiral conveyor into the next station.

A sensor monitors the presence of the caps in the spiral conveyor. Sensors monitor the infeed of caps for maximum and minimum backlog.



2035_05_01

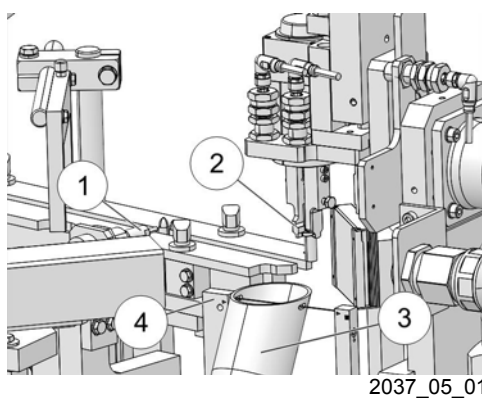
- Infeed gutter (1)
- Spiral conveyor (2)
- Sensor (3)
- Sensors (4) for minimum and maximum backlog

Station 2037 - Insert tip

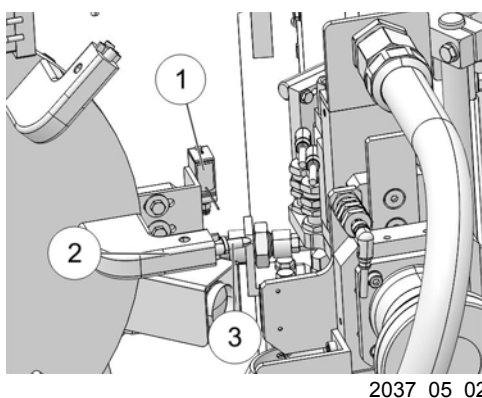
The separated tips are picked up by a gripper, rotated 90° after the camera check (station 2039) and fitted on the assembly wheel of station 2043. Bad parts are ejected via the bad parts chute.

A stopper reduces the backlog pressure in the infeed rail before separation.

The sensor ensures that a tip has been gripped.



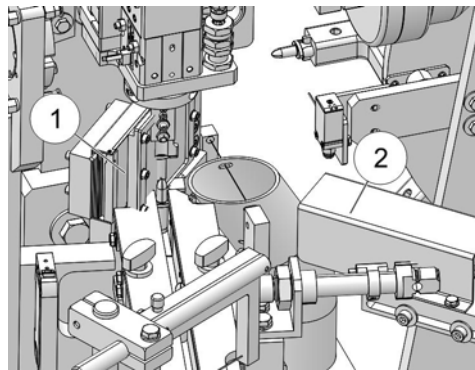
- Stopper (1)
- Gripper (2)
- Bad parts chute (3)
- "Bad part ejected" sensor (4)



- "Tip transferred" sensor (1)
- Tip receiver (2)
- Tip (3) in receiver

Station 2039 - Vision system tip

The tips are checked with a camera for size and completeness.



2039_05_01

- Background lighting (1)
- Camera (2)



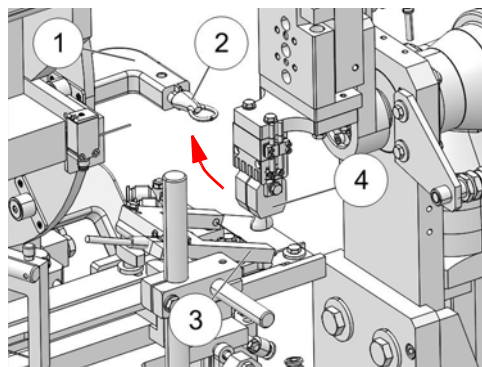
Observe!

For information about operation, function and adjustment, please refer to the manufacturer's operating instructions.

Station 2041 - Insert cap

A gripper picks up the caps from separation, swivels 90° and places them on the tips in the assembly wheel of station 2043.

A sensor (forked photoelectric sensor) checks the presence of the cap in the collecting position. A sensor checks whether the cap has been fitted on the tip



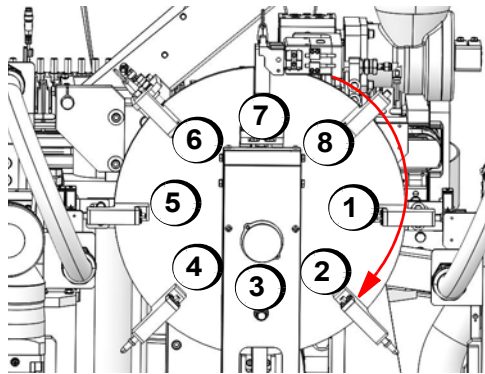
2041_05_01

The cap in the gripper is lifted off the infeed belt and fitted on the tip.

- Receiver (1)
- Fitted cap (2)
- "Cap presence" sensor (3)
- Presence sensor (4)

Station 2043 - Cap/tip assembly

4 stations are mounted on a vertical turntable. In addition there are 4 empty stations.

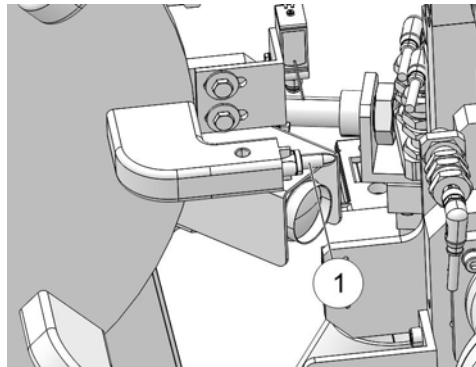


Turntable with 8 positions:

- Transfer station (1)
- Empty station (2)
- Empty station (3)
- Empty station (4)
- Cap/tip assembly station (5)
- Empty station (6)
- Cap/tip removal (7)
- Empty station (8)

2043_05_01

Tip transfer from
station 2037

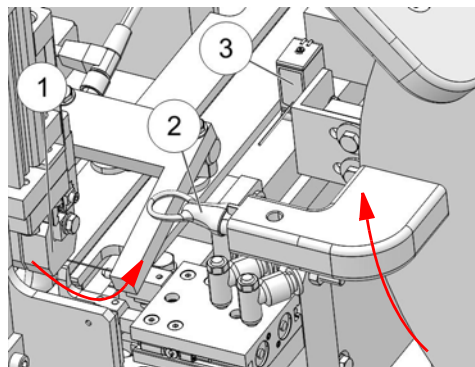


The tip is inserted in the 3 o'clock position with the opening pointing forwards.

- Tip (1)

2043_05_02

Assembly station

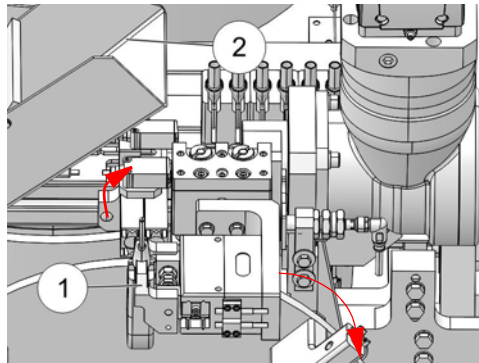


The cap is fitted on the tip in the 9 o'clock position.

- Caps infeed gripper (1)
- Cap on tip (2)
- Presence sensor (3)

2043_05_03

Discharge station



2043_05_04

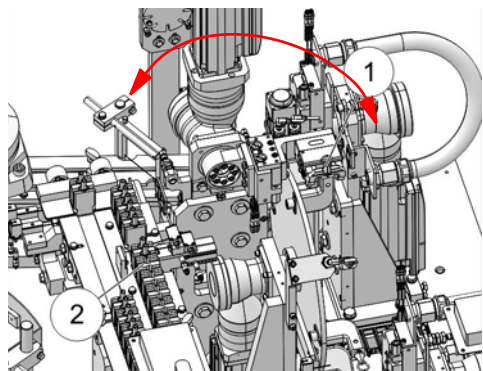
In the 12 o'clock position, the assembled caps/tips are taken from the turntable and placed in the pucks of the infeed belt.

A piston supports the process from underneath.

- Gripper (1)
- Bad parts chute (2)

Station 2045 - Cap/tip transfer

The assembled caps/tips are removed from the turntable by a gripper. The gripper rotates vertically 180° and places the assembled caps/tips on the pucks of the infeed belt.



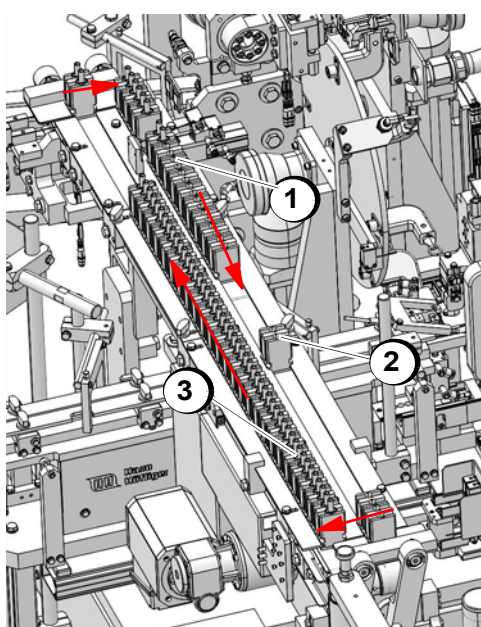
2045_05_01

- Gripper (1)
- Pucks (2)

Station 2047 - Load pucks/buffer Station 2048 - Unload pucks

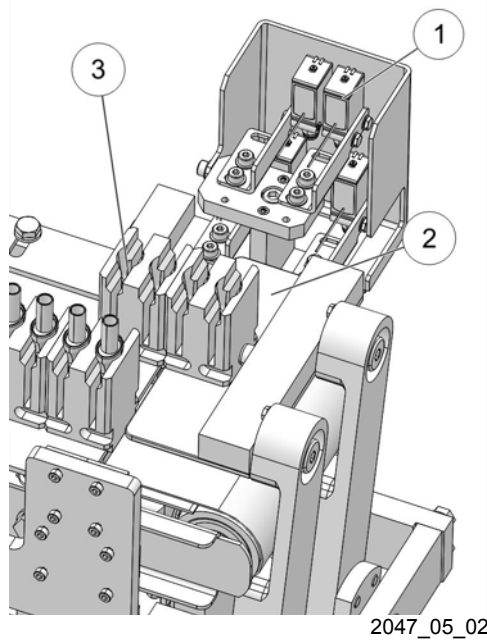
The infeed belt with pucks acts as a buffer with a capacity of approx. 2 production minutes. The empty pucks are transported back on the return conveyor belt. Lateral transport of the pucks is by means of a pneumatic pusher.

Sensors check insertion and pick-up of the caps/tips.

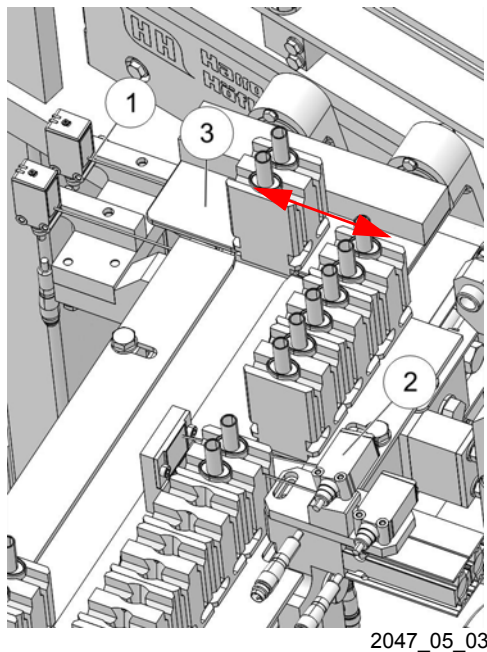


2047_05_01

- Infeed belt (1)
- Puck (2)
- Return conveyor belt (3)



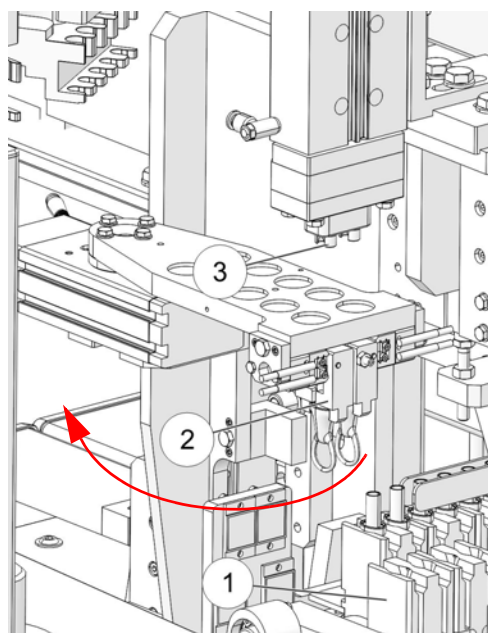
- Empty pucks presence sensor (1)
- Pneumatically actuated pusher (2)
- Empty pucks (3)



- Full pucks presence sensor (1)
- Pneumatically actuated pusher (2)
- Full pucks for unloading presence sensor (3)

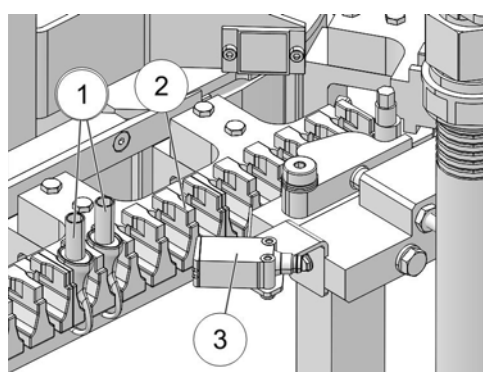
Station 2049 - Insert cap/tip

The gripper lifts two assembled caps/tips out of the pucks, rotates them horizontally 180° and places them in the cap/tip workpiece carrier of the oval runner. A sensor monitors successful transfer into the workpiece carrier.



2049_05_01

- Pucks (1)
- Gripper (2)
- Gripper (3)



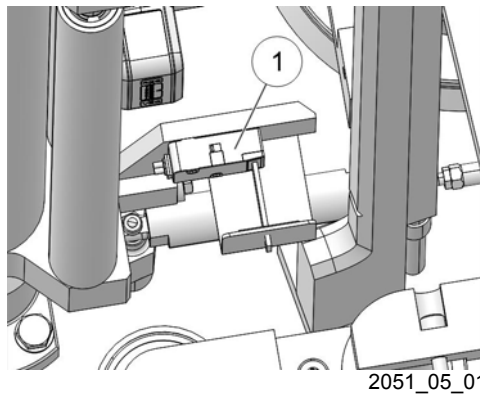
2049_05_02

- Cap/tip in oval runner (1)
- Cap/tip workpiece carrier in oval runner (2)
- Sensor (3)

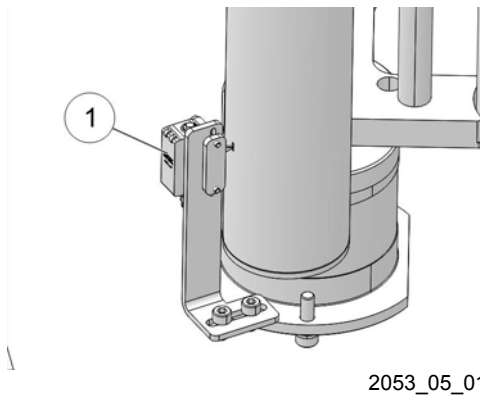
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Station 2051 - Position verification front film Station 2053 - Position verification rear film

The respective position of the film is checked in front of the sealing station with a laser sensor.



- Sensor for detection of film (1)

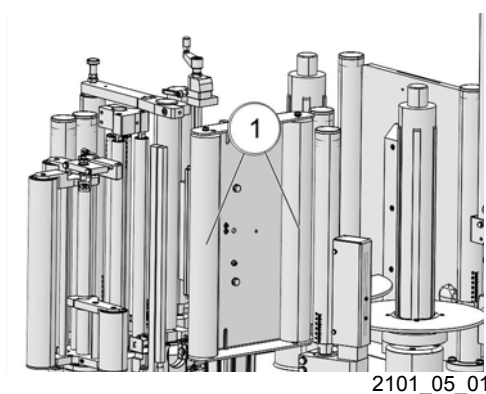


- Splice detection sensor (2)

Station 2101- Edge guiding front film Station 2103 - Edge guiding rear film

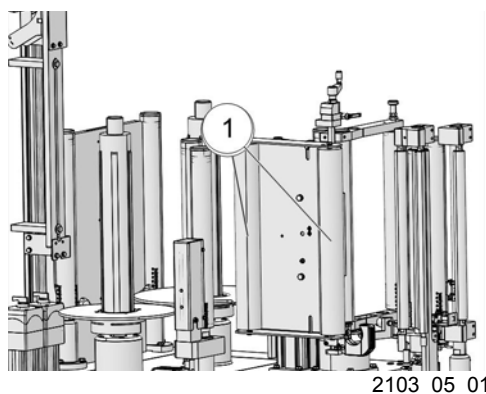
Web edge control adjusts any displacement of the web.

- By adjusting the axles of a pair of rollers (perpendicular to the web direction), the web path is adjusted.
- Sensors detect the edge of the web and send signals to make any corrections.



Web edge control station 2101 web:

- Adjustable roller pair (1)



Web edge control station 2103 web:

- Adjustable roller pair (1)

Station 2150 - Unwind front film Station 2151 - Unwind rear film

The station consists of:

- Servo-driven pneumatic expansion shaft
- Dancer
- Pneumatic splicing table
- Ionisation

The film is unwound constantly by the expansion shaft.

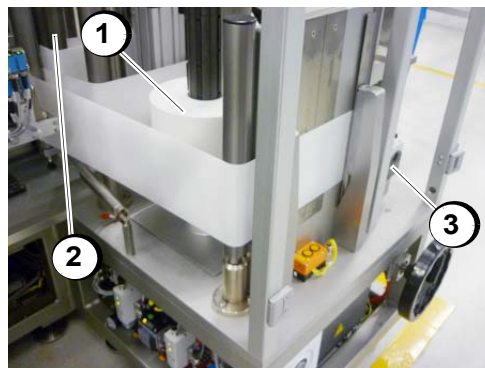
Both the adjustment of the required web tension as well as the control of the unwinding procedure are performed via a dancer control.

The station features a pneumatic splicing table.

The reel arbor can be swivelled manually into the horizontal position to allow the reel to be changed.

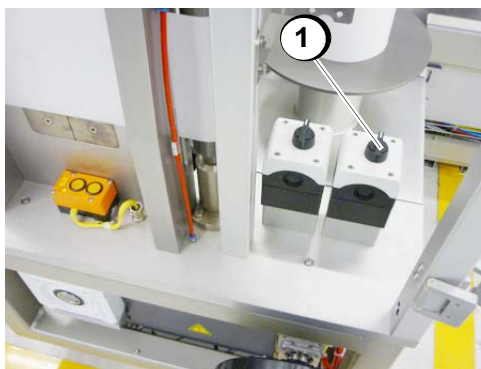
Unwinding with servo

- The pneumatic expansion shaft is mounted in safety chucks.
- Unwinding is directly driven by a servo motor with gear.
- The expansion shaft is tensioned using a switch.
- Specification of the unwinding speed based on the dancer position.
- Prewarning and detection of "reel end" by calculating the roll diameter.



e.g. unwinding station 2150:

- Unwinding (1)
- Dancer (2)
- Switch for tensioning (3)



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- Button (1) for tensioning and detensioning the reel winding arbor

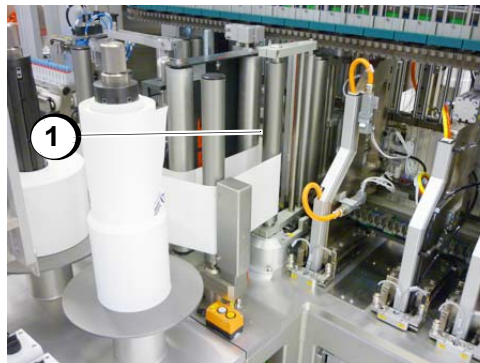
Dancer With the pendulum-mounted dancer arm, the unwinder located in front of it is controlled depending on position.

The web tension is generated pneumatically by a low-friction cylinder and can be manually adjusted via a pressure control valve on the HMI.

Position detection of the dancer arm takes place contactlessly by means of an analogue sensor.

Detection of "winding speed too fast / too slow" by comparing dancer position with winder drive.

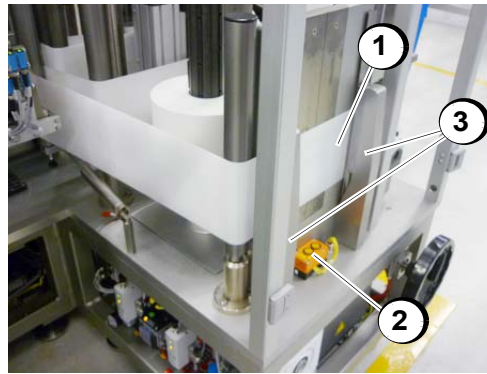
Detection of "film tear" using the dancer position.



- Dancer (1)

Splicing table The film is glued together on the splicing table (e.g. in the event of a film tear or a new film reel).

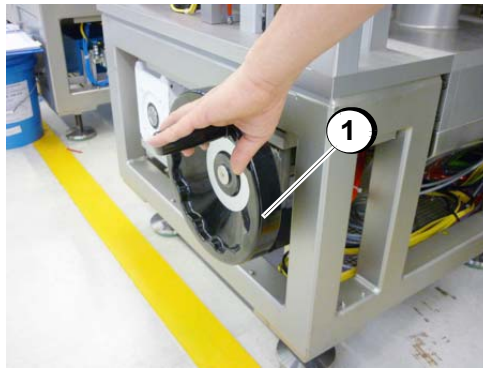
The film is affixed to the new reel during splicing by the pneumatic clamping bar. The buttons can be used to manually open and close the clamping bar



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- Splicing table (1)
- Button (2) for opening and closing the clamping bar (3)

Expansion shaft folded Expansion shaft folded into reel change position by handwheel



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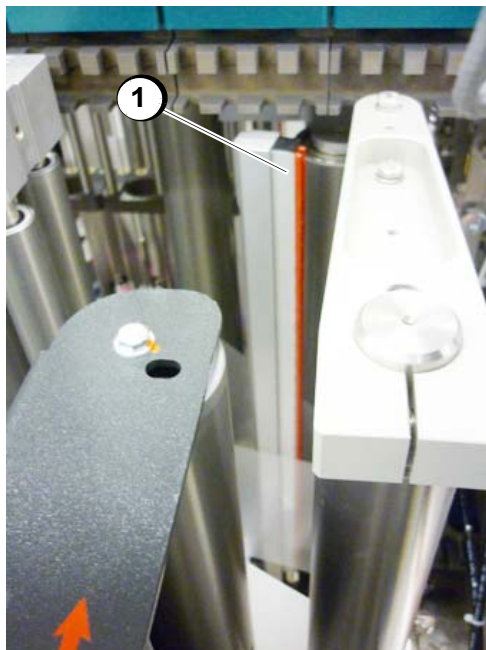
- Handwheel (1)



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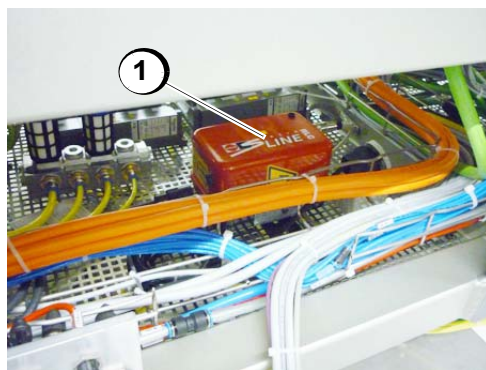
→ Expansion shaft folded 90°

Ionisation Ionisation discharges any electrostatic energy from the film.



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- Ionisation rod (1)



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- Mains unit (1)



Observe!

For information about operation, function and adjustment, please refer to the manufacturer's operating instructions.

Station 2170 - Rewind front film carrier Station 2171 - Rewind rear film carrier

The station consists of:

- Servo-driven pneumatic expansion shaft
- Dancer
- Pneumatic clamping jaws

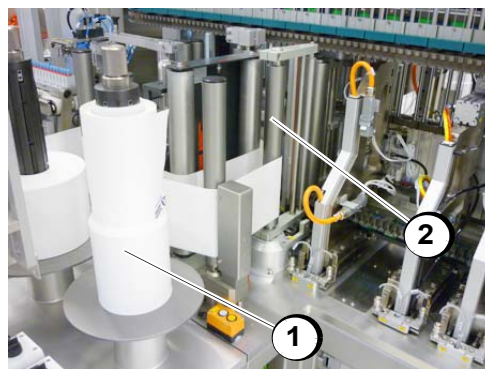
The film is constantly wound onto the expansion shaft.

Adjustment of the required web tension and also control of the unwinding procedure are performed via a dancer control.

The reel arbor can be swivelled manually into the horizontal position to allow the reel to be changed.

Winding with servo

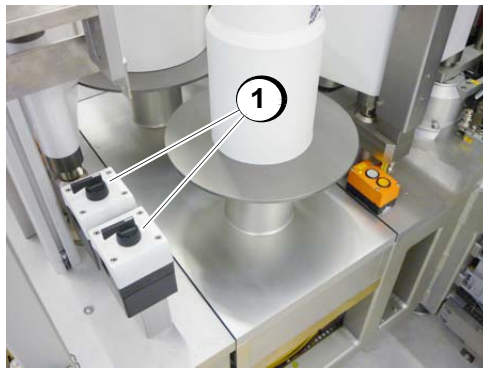
- The pneumatic expansion shaft is mounted in safety chucks.
- Winding is directly driven by a servo motor with gear.
- The expansion shaft is tensioned using a switch.
- Specification of the winding speed based on the dancer position.
- Prewarning and detection of "reel end" by calculating the roll diameter.



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e.g. Winding station 2170:

- Winding (1)
- Dancer (2)



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- Switch (1) for tensioning and detensioning the reel winding arbor

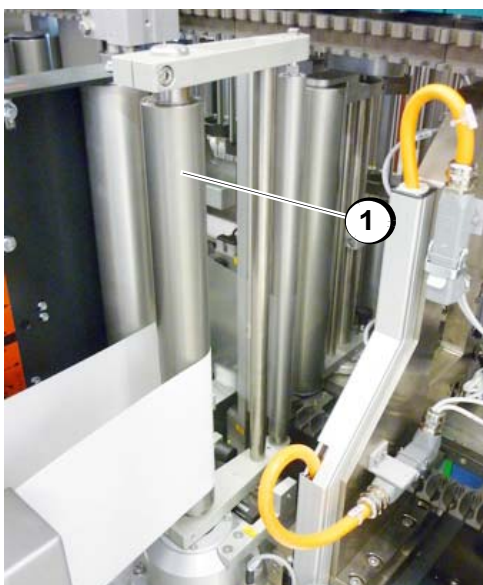
Dancer With the pendulum-mounted dancer arm, the unwinding which takes place in front of it is controlled depending on position.

The foil tension is generated pneumatically by a low-friction cylinder and can be manually adjusted via a pressure control valve.

Position detection of the dancer arm takes place contactlessly by means of an analogue sensor.

Detection of "winding speed too fast / too slow" by comparing dancer position with winder drive.

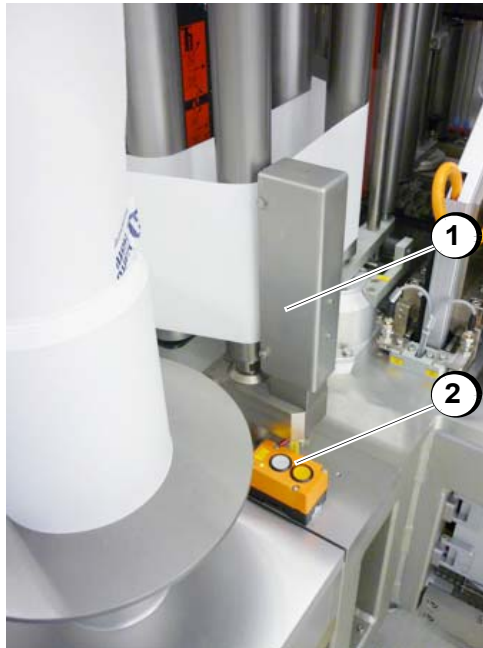
Detection of film tear using the dancer position.



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- Dancer (1)

Clamping jaws The clamping jaw fixes the film (e.g. in the event of a film tear or new film reel).
The clamping jaw can be opened and closed manually by pressing a button



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- Clamping jaw (1)
- Button (2) for opening and closing the clamping jaw